

Alexandria University
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CCE – Computer Vision
Assignment 4 Report

Pets Facial Expression Classification Using Deep Convolutional Neural Networks

1. Introduction

Understanding animal emotions through facial expressions has become an emerging research direction in computer vision, inspired by advances in human facial expression recognition. Automatic recognition of pets' emotions can contribute to better animal welfare, enhanced interaction between pets and owners, and improved monitoring of animal health and behavior.

This report presents a comprehensive experimental study on pets facial expression classification using deep convolutional neural networks. Multiple well-known CNN architectures were implemented and trained from scratch, followed by a transfer learning experiment using ImageNet-pretrained weights. The models were evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrices.

2. Dataset Description

The Pets Facial Expression Dataset consists of 1000 RGB images of various animals, including dogs, cats, rabbits, hamsters, sheep, horses, and birds. The dataset captures different emotional expressions across four classes:

- Angry
- Other (neutral or ambiguous expressions)
- Sad
- Happy

Dataset statistics:

- Total number of images: 1000
- Number of classes: 4
- Images per class (test set):
 - Angry: 49
 - Other: 38
 - Sad: 59
 - Happy: 54

The dataset was split into training and validation sets using an 80% / 20% ratio.

3. Data Preprocessing and Augmentation

All images were resized to a fixed input resolution compatible with the selected CNN architectures. Pixel values were normalized to improve numerical stability during training.

To reduce overfitting and improve generalization, data augmentation was applied during training. Augmentation techniques included random horizontal flipping, random rotations, and slight scaling variations. This was particularly important given the limited dataset size.

4. Models Trained from Scratch

Five well-known CNN architectures were implemented and trained from scratch using randomly initialized weights:

- VGG
- ResNet
- MobileNet
- InceptionNet
- DenseNet

Each model was trained using the same dataset split, loss function (cross-entropy), optimizer, and early stopping strategy based on validation loss.

a. VGG Network

The VGG model contained approximately 134 million trainable parameters, making it the largest and most computationally expensive architecture in this study.

Training behavior:

- The model exhibited very slow convergence.
- Severe overfitting and poor generalization were observed.
- Early stopping was triggered after 12 epochs.

Final performance metrics:

- Accuracy: 0.2000
- Precision: 0.2169
- Recall: 0.2520
- F1-score: 0.1280

The confusion matrix showed that the model was heavily biased toward predicting the “Other” class, failing to correctly distinguish between emotional expressions. This indicates that VGG is not suitable for training from scratch on small datasets due to its extremely high parameter count.

b. ResNet

The ResNet architecture used residual connections to mitigate the vanishing gradient problem. It contained approximately 11.2 million trainable parameters.

Training behavior:

- Faster convergence compared to VGG.
- Improved stability during training.
- Moderate overfitting after extended training.

Final performance metrics:

- Accuracy: 0.5200
- Precision: 0.5440
- Recall: 0.5232
- F1-score: 0.5185

The confusion matrix demonstrated balanced performance across all classes, with particularly strong precision for the Angry class. Residual learning significantly improved performance compared to VGG.

c. MobileNet

MobileNet is a lightweight architecture designed for efficiency, using depthwise separable convolutions. It contained approximately 3.2 million trainable parameters.

Training behavior:

- Rapid convergence.
- Lower computational cost.
- Slightly reduced representational capacity.

Final performance metrics:

- Accuracy: 0.4300
- Precision: 0.4613
- Recall: 0.4373
- F1-score: 0.4390

MobileNet showed reasonable performance given its low parameter count but struggled to distinguish subtle emotional differences, particularly for the Sad class.

d. InceptionNet

InceptionNet employed parallel convolutional paths to capture multi-scale features, with approximately 4.5 million parameters.

Training behavior:

- Stable training.
- Moderate convergence speed.
- Balanced feature extraction.

Final performance metrics:

- Accuracy: 0.4200
- Precision: 0.4519
- Recall: 0.4340
- F1-score: 0.4124

The confusion matrix revealed strong recall for Angry expressions but weaker performance for Sad expressions, suggesting sensitivity to certain facial features.

e. DenseNet (From Scratch)

DenseNet introduced dense connectivity between layers, improving feature reuse and gradient flow. It had approximately 1.82 million trainable parameters.

Training behavior:

- Smooth and stable convergence.
- Reduced overfitting.
- Efficient parameter utilization.

Final performance metrics:

- Accuracy: 0.5450
- Precision: 0.5602
- Recall: 0.5482
- F1-score: 0.5340

DenseNet achieved the best overall performance among all models trained from scratch, with strong results across all classes.

5. Comparative Analysis of Scratch Models

A comparison of all models trained from scratch is summarized below:

Model	Accuracy	Precision	Recall	F1-score
VGG	0.2000	0.2169	0.2520	0.1280
ResNet	0.5200	0.5440	0.5232	0.5185
MobileNet	0.4300	0.4613	0.4373	0.4390
InceptionNet	0.4200	0.4519	0.4340	0.4124
DenseNet	0.5450	0.5602	0.5482	0.5340

DenseNet was selected as the best-performing architecture for transfer learning.

6. Transfer Learning with DenseNet

To further enhance performance, DenseNet121 pretrained on ImageNet was used. Transfer learning was performed in two stages:

Stage 1: Training the classifier head while freezing the backbone.

Stage 2: Fine-tuning the entire network.

The pretrained DenseNet contained approximately 6.96 million trainable parameters.

6.1 Stage 1 Results

- Best validation accuracy: 0.6250
- Significant improvement over scratch training.

6.2 Stage 2 Results (Fine-Tuning)

Fine-tuning led to dramatic performance gains:

- Accuracy: 0.9500
- Precision: 0.9503
- Recall: 0.9521
- F1-score: 0.9504

The confusion matrix showed near-perfect classification across all four classes, with particularly strong recall for Sad and Other expressions.

7. Scratch vs Transfer Learning Comparison

Training Method	Accuracy	Precision	Recall	F1-score
From Scratch	0.5450	0.5602	0.5482	0.5340
Transfer Learning	0.9500	0.9503	0.9521	0.9504

Transfer learning improved accuracy by approximately 40.5 percentage points, highlighting the importance of pretrained representations when working with limited datasets.

8. Discussion

The experimental results clearly demonstrate that:

- Large models like VGG are unsuitable for small datasets when trained from scratch.
- Architectures with efficient connectivity (ResNet and DenseNet) perform significantly better.
- DenseNet achieves the best balance between parameter efficiency and accuracy.
- Transfer learning dramatically improves generalization and convergence speed.

The use of ImageNet-pretrained weights allowed the network to leverage rich low-level and mid-level features, leading to near-perfect classification performance.

9. Conclusion

This report presented a detailed comparison of multiple CNN architectures for pets facial expression classification. DenseNet achieved the best performance when trained from scratch, while transfer learning using ImageNet-pretrained DenseNet resulted in a substantial performance boost, achieving 95% classification accuracy.

The results confirm that transfer learning is essential for high-performance computer vision systems when training data is limited.